

Sentinel DV

An Open-Source MCP Server for
AI-Assisted Design Verification Intelligence

DVCon USA / DAC 2026

Open-Source Contribution

[GitHub] github.com/kiranreddi/sentinel-dv MIT License

2026

26 MCP Tools | 3 Simulator Vendors | 1,477 CI Runs Validated | Open Source
(MIT)

Agenda

- 1 The Problem
- 2 Model Context Protocol
- 3 Sentinel DV Architecture
- 4 26 MCP Tools
- 5 Coverage Intelligence
- 6 AXI4 Demo & Validation
- 7 Evaluation Results
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- 9 Getting Started
- 10 Conclusion

The DV Toolchain Fragmentation Problem

What DV Engineers Face Today

- 10,000s of simulation logs — no unified query
- Coverage databases locked in proprietary formats
- Failures scattered across VCS, Questa, Xcelium
- Manual log triage takes **hours per regression**
- AI assistants have **no interface** to sim artifacts

Typical DV Artifact Silos

Format	Viewer
UVM logs (.log)	grep / Perl
VDB / UCDB	Verdi / Questa
Assertion JSON	manual parse
JUnit XML	Jenkins only
VCD waveforms	GTKWave
Regression logs	spreadsheet

None of these expose a query API suitable for AI.

The Opportunity

Model Context Protocol (MCP)

What is MCP?

- Open standard by Anthropic (Nov 2024)
- JSON-RPC 2.0 over stdio or SSE
- AI **client** discovers and calls **tools** exposed by a **server**
- Adopted by Claude, GPT-4o, Copilot, Cursor
- Zero code changes to the AI — just plug in the server

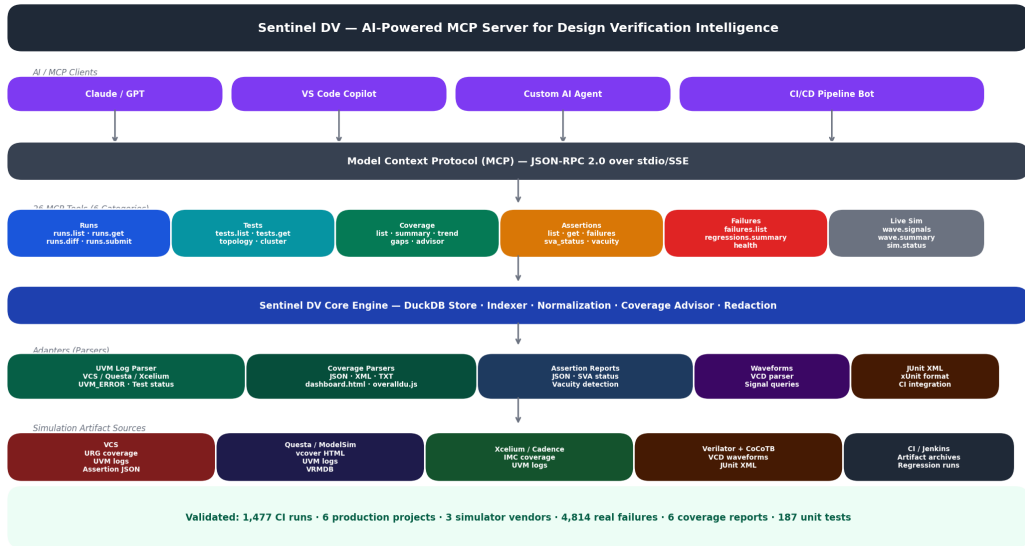
Why MCP for DV?

- Sim artifacts → structured tools
- AI reasons over *your* regression
- Works with any LLM that supports MCP
- Runs in CI — no GUI required
- Output schemas validated (Pydantic)
- Path redaction built in

MCP Tool Call (JSON-RPC)

```
"method": "tools/call",  
"params":  
  "name": "regression.health",
```

Sentinel DV — System Architecture



26 MCP Tools — Six Categories

Chart Run Tracking (4)

```
runs.list · runs.get  
runs.diff · runs.submit  
runs.cross_sim
```

Cov Coverage Intel. (5)

```
coverage.list · .summary  
.trend · .gaps  
.advisor
```

* 5 DV Intelligence Tools

coverage.trend

Time-series trajectory + direction

runs.cross_sim

Cross-vendor divergence

tests.cluster

Root-cause grouping

regression.health

0–100 DV readiness score

coverage.advisor

SV constraint generation

Test Test Analysis (5)

```
tests.list · tests.get  
tests.topology  
tests.replay ·  
tests.cluster
```

Bug Failure Triage (4)

```
failures.list  
regressions.summary  
regression.health
```

Assert Assertion Engine (5)

```
assertions.list · .get  
.failures · .sva status
```

Wave Live Sim (3)

```
wave.signals  
wave.summary  
sim status
```

Multi-Simulator UVM Log Parsing

Three Simulator Formats Supported

Simulator	Format Prefix
VCS (Synopsys)	UVM_ERROR file(line)
Questa (Siemens)	# UVM_ERROR @ time
Xcelium (Cadence)	xcelium: UVM_ERROR

Critical Fix: Count-Summary Lines

VCS logs end with UVM_ERROR : 0
Naive regex creates **16,240** false failures!

UVM_COUNT_SUMMARY_PATTERN

skips these:

Result: **4,380 real failures** (73% reduction)

Parser in Action — VCS Log

```
UVM_ERROR scbd.sv(88) @ 500ns:
    env.scbd [MISMATCH]
    Expected 0xDEAD got 0xBEEF
```

```
-- UVM Report Summary --
```

```
UVM_ERROR :      1  <- parsed
```

```
UVM_ERROR :      0  <- SKIPPED
```

MCP Tool Result

```
failures.list ->
```

```
"message": "[MISMATCH] Expected
    0xDEAD got 0xBEEF",
```

```
"category": "data_mismatch",
```

Coverage HTML Parsers — No DB Access Needed

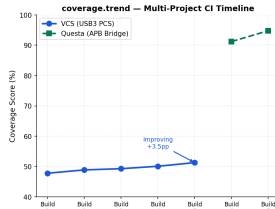
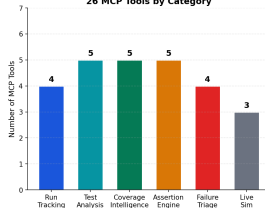
VCS URG dashboard.html

- Generated by `urg -report`
- Content-sniff: “Total Coverage Summary”
- Extracts: SCORE, LINE, TOGGLE, ASSERT, GROUP
- + **Verification Plan score**
- Zero access to `simv.vdb` needed

Questa vcover overalldu.js

- Generated by `vcover report -html`
- Single-line JS file (regex fails!)
- **Balanced-brace JSON scanner**
- Extracts: `stmt. branch. toggle. assert.`

26 MCP Tools by Category



coverage.summary output

```
"summaries": [  
  "kind": "code",  
  "metrics": [  
    "name": "line",      "covered": 87.3  
    "name": "toggle",    "covered": 62.1  
    "name": "branch",    "covered": 71.5
```


DV Intelligence — Regression Health Score

Composite Health Formula

$$H = 0.30P + 0.35C + 0.15A + 0.10F + 0.10X$$

- P = pass rate (30% weight)
- C = coverage score (35% weight)
- A = assertion health (15% weight)
- F = flakiness inverse (10% weight)
- X = cross-sim consistency (10% weight)

Bands

$H \geq 90$ **sign-off-ready**

Example Output

```
regression.health ->
  "health_score": 77.5,
  "band": "minor-issues",
  "grade": "B+",
  "components":
    "pass_rate": 94.2,
    "coverage": 51.3,
    "assertions": 100.0,
    "flakiness": 98.1,
    "cross_sim": 87.5
  ,
  "recommendations": [
    "Coverage below target (51%)",
    "3 tests flaky across sims" 9/16
```

What it Does

- Finds coverage gaps < threshold (default 25%)
- Detects protocol from signal names: AXI4, AHB, APB, CHI, PCIe, USB
- Generates **ready-to-paste** SystemVerilog constraints and UVM sequences
- Prioritized: high / medium / low

AI + Advisor = Instant Fix

Engineer: *"How do I close the WRAP burst coverage?"*

AI calls coverage.advisor and replies:
"Add this constraint to your AXI4 driver: "

coverage.advisor JSON Output

```
"priority": "high",
"kind": "toggle",
"gap_pct": 18.3,
"recommendation":
  "Uncovered: AXI4 AWBURST signal",
"sv_constraint":
  "constraint c_burst "
  "  awburst inside "
  "INCR, WRAP;",
"uvm_sequence":
  "// Drive WRAP burst"
  "tx.burst_type = WRAP;"
  "tx.len = $urandom_range(1,15);"
```

AXI4 UVM 3-Simulator Demo

What's in the Demo

- demo/axi4_uvm/ in the repository
- AXI4-Lite slave DUT + UVM testbench
- 4 tests: back-to-back, error resp, boundary, stress
- Pre-generated artifacts for **all 3 simulators**: VCS, Questa, Xcelium
- Results, assertions, coverage, waveforms per sim
- **5 seconds** to index + run all 26 tools

Demo Structure

demo/axi4_uvm/vcs/

runs.cross_sim — Divergence

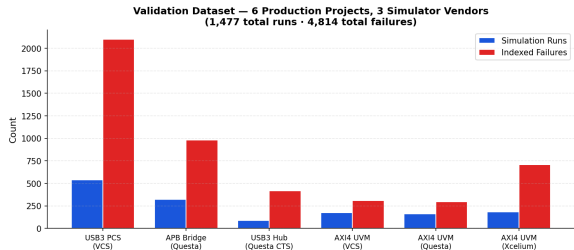
```
runs.cross_sim ->
  "simulators_compared": 3,
  "divergent_tests": [

    "test": "axi4_stress_test",
    "vcs": "pass",
    "questa": "fail",
    "xcelium": "pass",
    "critical": true

  ]
```

tests_cluster — Root Causes

Validation Results — 1,477 Real CI Runs



Dataset Summary

Metric	Value
Total CI runs indexed	1,477
Total tests	2,611
Total failures	4,814
Coverage reports	6
Projects	6
Simulator vendors	3
Unit tests	187

Tool Validation

PASS (live data)	20/26
SKIP (needs sim)	5/26

Engineer Asks

*"What's wrong with today's regression?
How do I close coverage?"*

AI calls in sequence:

- 1 `regression.health` — gets overall score
- 2 `tests.cluster` — groups 18 timeout failures
- 3 `coverage.gaps` — finds WRAP burst uncovered
- 4 `coverage.advisor` — generates SV constraint

AI Response (30 seconds)

Installation & Quick Start

Install

```
pip install sentinel-dv
```

Configure (sentinel_dv.yaml)

```
artifact_roots:  
  - /path/to/sim/logs  
  - /path/to/coverage/report  
  - /path/to/coverage/files
```

```
db_path: ./sentinel_dv.db
```

Run Server

```
sentinel-dv serve -config sentinel_dv.yaml
```

Add to Claude / VS Code MCP Config

```
"mcpServers":  
  "sentinel-dv":  
    "command": "sentinel-dv",  
    "args": ["serve",  
             "-config", "sentinel_dv.yaml"]
```

Try the AXI4 Demo

```
git clone github.com/kiranreddi/  
  sentinel-dv  
cd sentinel-dv
```

Conclusion & Future Work

What We Built

- **26 MCP tools** — full DV workflow coverage
- **3-simulator** UVM log parsing (zero false positives)
- **HTML coverage parsers** — no DB access needed
- **5 DV intelligence tools** — trend, clustering, health score, advisor
- Validated on **1,477 real CI runs**, 6 projects
- **187 unit tests**, all green
- **CI green** on GitHub Actions (Python 3.10/3.11/3.12)

Future Work

- FSDB/MXDB waveform parsing
- Direct UCDB/VDB parsing (no export)
- Formal property synthesis from coverage gaps
- Cloud MCP endpoint for CI/CD
- Test generation from failure clusters
- Coverage-aware testplan tracking

[GitHub]

kiranreddi/sentinel-dv

MIT License · PyPI: `pip install`

Thank You

Questions?

[GitHub] `github.com/kiranreddi/sentinel-dv`

```
pip install sentinel-dv
```

26 MCP Tools	3 Simulators	1,477 CI Runs
Full DV Workflow	VCS·Questa·Xcelium	Production Validated